BEM abundance estimates pooling across time periods

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Introduction

In this document, I aimed to pool the distance data (of clusters) from the three time periods (2000, 2007, 2015) to estimate the abundance of Black-eared Miner clusters and individuals. The purpose of pooling the data was to increase the sample size and increase precision of abundance estimates.

Load packages and data

```
library(Distance)
library(dplyr)
library(tidyr)

# read data
data.2015 <- read.csv("Data/data.frame.2015b.csv", header = TRUE)
data.2007 <- read.csv("Data/data.frame.2007.csv", header = TRUE)
data.2000 <- read.csv("Data/data.frame.2000e.csv", header = TRUE)

# prepare data
data.2000 <- data.2000 %>%
  mutate(Region.Label =
           case_when(Region.Label == "Bookmark" ~ as.factor(2000)))
data.2007 <- data.2007 %>%
  mutate(Region.Label =
           case_when(Region.Label == "Bookmark" ~ as.factor(2007)) )
data.2015 <- data.frame(Region.Label = rep(as.factor(2015), nrow(data.2015)),
                        Area = data.2015$Area,
```

```
                        Sample.Label = data.2015$Sample.Label,
                        Effort = data.2015$Effort,
                        distance = data.2015$distance,
                        size = data.2015$size)

data.pooled <- rbind(data.2000, data.2015) # left out 2007 because of inconsistent samplin

data.pooled <- data.pooled %>%
  mutate(size = if_else(size < 6, NA_real_, as.numeric(size)),
         distance = if_else(is.na(size), NA_real_, as.numeric(distance)),
         Area = 900)

nrow(data.pooled[!is.na(data.pooled$distance),]) # how many datapoints
```

```
[1] 103
```

I only use the data from 2000 and 2015 because the sampling protocols are similar (i.e., entire 900 sq km sampled, instead of only unburnt areas sampled in 2017). I also ignore geographical stratification (i.e., OLD vs. BURNT) because `ds()` cannot accomodate changes in stratification between time periods (i.e., `Area` has to be consistent within each strata).

Instead, I treat `Year` as the stratification factor to allow year-specific estimates of abundance later on. Assumptions associated with this decision are: (i)

Explore the data

```
# plot
par(mfrow = c(2,2))
hist(data.pooled$distance, breaks = seq(0,300,10), main="", xlab="Distance (m)")
hist(data.pooled$distance, breaks = seq(0,300,20), main="", xlab="Distance (m)")
hist(data.pooled$distance, breaks = seq(0,300,30), main="", xlab="Distance (m)")
hist(data.pooled$distance, breaks = seq(0,300,50), main="", xlab="Distance (m)")
```

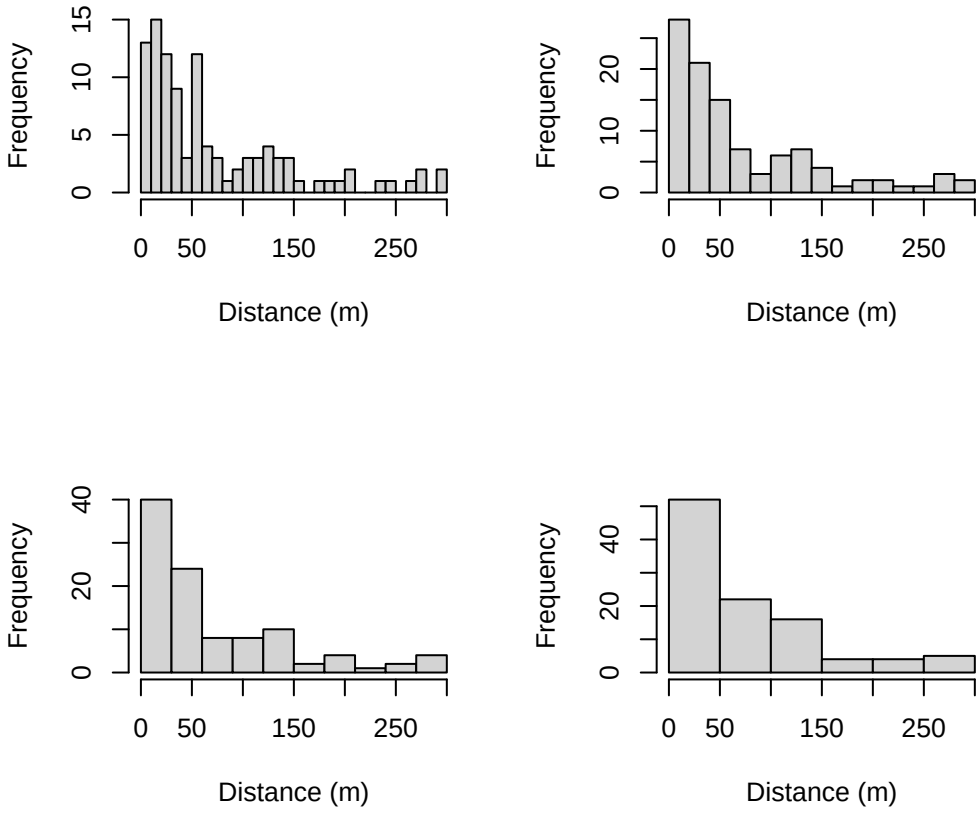


Figure 1: Frequency distribution of distances to nest or largest group.

Distance analysis

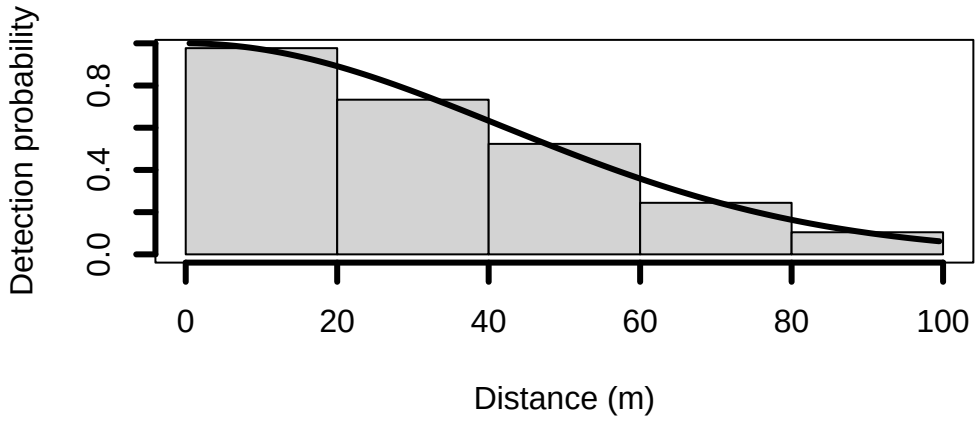
I select a binning interval of 20m, with right-truncation at 100m to avoid non-monotonicity in the detection fit…

I first fit a model with year treated as a stratification factor (i.e., under `Region.Label`). By combining years, detection functions fitted for all years are the same key function, but the scale parameter of the key function differs between years.

```
conversion <- convert_units("meter", "meter", "square kilometer")
mybreaks <- seq(0,100,20)
```

```
# half-normal
hn_strat <- ds(data.pooled,
                truncation = list(left=0, right=100),
               convert_units = conversion, cutpoints = mybreaks,
               key = "hn", adjustment = "cos", #order = 2,
               formula=~as.factor(Region.Label))

# check fit
plot(hn_strat, main="", xlab="Distance (m)",
     showpoints=FALSE, lwd=3, xlim=c(0, 100),pdf=F)
```



Examining the `summary()` output, we see the year-specific estimates being provided:

```
summary(hn_strat)
```

```
Summary for distance analysis
Number of observations :  74
Distance range         :  0  -  100

Model : Half-normal key function
AIC   : 211.2373

Detection function parameters
```

```
Scale coefficient(s):
                               estimate        se
(Intercept)                   3.7996087 0.1591049
as.factor(Region.Label)2015 -0.1248315 0.2179154

                        Estimate          SE         CV
Average p              0.5165209  0.04975035 0.09631818
N in covered region 143.2662216 18.02653185 0.12582542

Summary for clusters

Summary statistics:
  Region Area CoveredArea  Effort  n    k           ER        se.ER     cv.ER
1   2000  900        96.6  483000 38  483 7.867495e-05 1.226312e-05 0.1558708
2   2015  900       117.9  589500 36  590 6.106870e-05 9.870672e-06 0.1616323
3  Total 1800       214.5 1072500 74 1073 6.987183e-05 7.871061e-06 0.1126500

Abundance:
  Label  Estimate       se        cv      lcl       ucl       df
1  2000  648.5376 133.8516 0.2063900 433.9108  969.3260 308.7721
2  2015  562.2635 119.0114 0.2116648 372.4719  848.7627 334.3336
3 Total 1210.8011 179.1088 0.1479259 906.5879 1617.0955 354.6123

Density:
  Label  Estimate         se        cv       lcl       ucl       df
1  2000 0.7205974 0.14872406 0.2063900 0.4821232 1.0770289 308.7721
2  2015 0.6247372 0.13223489 0.2116648 0.4138576 0.9430697 334.3336
3 Total 0.6726673 0.09950491 0.1479259 0.5036600 0.8983864 354.6123

Summary for individuals

Summary statistics:
  Region Area CoveredArea  Effort   n    k           ER        se.ER     cv.ER
1   2000  900        96.6  483000 490  483 0.0010144928 0.0001826765 0.1800668
2   2015  900       117.9  589500 377  590 0.0006395250 0.0001110958 0.1737161
3  Total 1800       214.5 1072500 867 1073 0.0008270089 0.0001069030 0.1292646
  mean.size   se.mean
1  12.89474 1.1769415
2  10.47222 0.6755199
3  11.71622 0.6978892

Abundance:
  Label  Estimate       se        cv       lcl       ucl       df
```

```
1  2000  8362.722 1883.473 0.2252225  5400.002 12950.943 376.5625
2  2015  5888.148 1301.458 0.2210301  3832.412  9046.598 373.4536
3 Total 14250.870 2289.381 0.1606485 10413.155 19502.955 428.4065

Density:
  Label Estimate       se        cv      lcl      ucl       df
1  2000 9.291913 2.092748 0.2252225 6.000002 14.38994 376.5625
2  2015 6.542387 1.446064 0.2210301 4.258235 10.05178 373.4536
3 Total 7.917150 1.271878 0.1606485 5.785086 10.83497 428.4065

Expected cluster size
  Region Expected.S se.Expected.S cv.Expected.S
1   2000   12.89474     1.1625563    0.09015743
2   2015   10.47222     0.6666368    0.06365763
3  Total   11.76979     0.7178546    0.06099131
```

The summary also shows beta parameters associated with year-specific estimates of the shape parameter of the half-normal key function. The standard errors of the beta (for 2015) is larger than the estimate itself. This suggests there us not a strong 'year' signal on the shape of the detection functions. Perhaps a model with a pooled detection function would be equally suitable for these data. See reference here.

```
hn_pooled <- ds(data.pooled,
                truncation = list(left=0, right=100),
               convert_units = conversion, cutpoints = mybreaks,
               key = "hn", adjustment = "cos")
```

```
Warning in create_bins(data, cutpoints): Some distances were outside bins and
have been removed.

Starting AIC adjustment term selection.

Fitting half-normal key function

AIC= 209.572

Fitting half-normal key function with cosine(2) adjustments

AIC= 211.572
```

```
Half-normal key function selected.
```

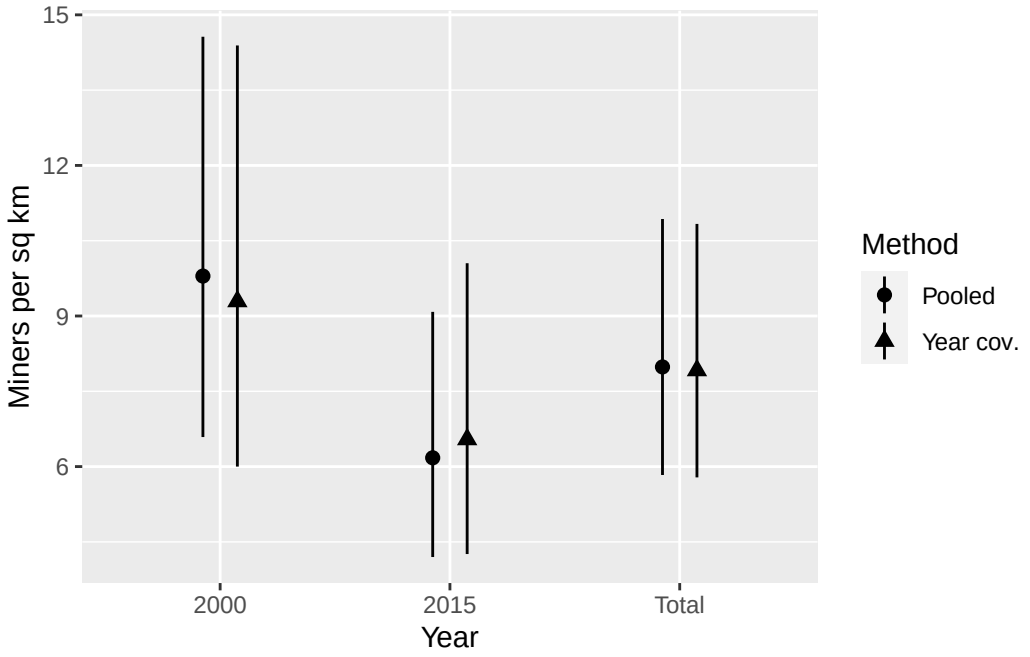
Having a look at the point and interval estimates of the two models bears out the minute difference between the models.

```
dens_yrcov <- hn_strat$dht$individuals$D
dens_cons <- hn_pooled$dht$individuals$D

compare_est <- rbind( data.frame(year = dens_yrcov$Label,
                            estimate = dens_yrcov$Estimate,
                            lcl = dens_yrcov$lcl,
                            ucl = dens_yrcov$ucl,
                            Method = "Year cov."),
                      data.frame(year = dens_cons$Label,
                            estimate = dens_cons$Estimate,
                            lcl = dens_cons$lcl,
                            ucl = dens_cons$ucl,
                            Method = "Pooled") )

library(ggplot2)

ggplot(compare_est, aes(x=year, y=estimate, shape = Method)) +
  geom_pointrange(aes(ymin=lcl, ymax=ucl), color="black",
                  position=position_dodge(width=0.3)) +
  labs(x="Year", y = "Miners per sq km")
```



We can go with the pooled model, which has a slightly smaller year-specific CV($\hat{D}$); i.e., higher precision.

Comparing the 2015 density estimates with that of the previous document ("1_Nest-vs-First.pdf"), we see similar point estimates (6.18 vs 6.25) and improves to the CV (0.20 vs 0.28). The improvement in CV likely comes from the better detection model fit to the data (with a truncation distance of 100m and binning distance of 20m) AND some pooling effect.